OLAP CUBE REPORT

Sales Analysis by Branch and Product Line

Data Warehouse and Business Intelligence Analysis

Prepared from the supplied Jupyter Notebook

1. Introduction

Online Analytical Processing (OLAP) is used to analyse data from multiple business perspectives. In this analysis, an OLAP cube was created from the sales fact table to evaluate total sales across two major dimensions: Branch and Product Line. The cube provides a compact multidimensional view that supports comparison of sales performance between branches and product categories.

2. Data Warehouse Structure

The notebook organises the cleaned supermarket sales data into a star-schema style structure. The central FactSales table contains 1,000 sales records and 17 attributes. It is connected to five dimension tables: DimDate, DimBranch, DimProduct, DimCustomer, and DimPayment.

Table	Size	Purpose
FactSales	1,000 × 17	Central fact table containing sales measures and foreign keys
DimDate	89 × 9	Date-related attributes
DimBranch	3 × 3	Branch and city information
DimProduct	6 × 2	Product line information
DimCustomer	4 × 3	Customer type and gender information
DimPayment	3 × 2	Payment method information

3. OLAP Cube Design

The OLAP cube was implemented using a Pandas pivot table. The measure selected for aggregation was Sales, while BranchKey was used as the row dimension and ProductKey as the column dimension. The aggregation function was SUM, which calculates the total sales value for every Branch–Product Line combination. Dimension keys were then mapped back to their descriptive branch and product-line names to make the cube readable.

- **Measure:** Sales
- **Row Dimension:** Branch
- **Column Dimension:** Product Line
- **Aggregation:** SUM of Sales
- **Number of Branches:** 3 — Alex, Giza, Cairo
- **Number of Product Lines:** 6

4. OLAP Cube Results

The resulting cube shows total sales for each product line within each branch. Values below are taken directly from the OLAP output in the notebook.

4.1 Total Sales by Branch and Product Line

Branch	Health and beauty	Electronic accessories	Home and lifestyle	Sports and travel	Food and beverages	Fashion accessories	Branch Total
Alex	12,597.75	18,317.11	22,417.20	19,372.70	17,163.10	16,332.51	106,200.37
Giza	16,615.33	18,968.97	13,895.55	15,761.93	23,766.85	21,560.07	110,568.71
Cairo	19,980.66	17,051.44	17,549.16	19,988.20	15,214.89	16,413.32	106,197.67

5. Analysis and Key Findings

The cube reveals meaningful differences in sales performance across branches and product lines. Giza records the highest overall branch sales at 110,568.71. Alex follows with 106,200.37, while Cairo records 106,197.67. Therefore, Giza is the strongest branch in terms of total sales, although the overall performance of all three branches is relatively close.

Across product lines, Food and beverages produces the highest combined sales value at 56,144.84. Sports and travel is second with 55,122.83, followed closely by Electronic accessories at 54,337.53 and Fashion accessories at 54,305.90. Health and beauty has the lowest combined total at 49,193.74.

The highest individual Branch–Product Line value is Giza – Food and beverages with total sales of 23,766.86. The lowest value is Alex – Health and beauty with 12,597.75. This shows that the contribution of a product category can vary considerably between branches.

5.1 Product Line Totals

Rank	Product Line	Total Sales
1	Food and beverages	56,144.84
2	Sports and travel	55,122.83
3	Electronic accessories	54,337.53
4	Fashion accessories	54,305.90
5	Home and lifestyle	53,861.91
6	Health and beauty	49,193.74

6. OLAP Operations Supported by the Cube

Although the notebook creates the cube primarily as a pivot-table summary, the same structure supports common OLAP operations. Slice can isolate a single branch or product line; dice can filter multiple branches and product categories; roll-up can aggregate detailed cube values into branch totals or product-line totals; drill-down can move from summarized values to detailed fact records; and pivot can rotate dimensions to view the analysis from another perspective.

7. Business Interpretation

The cube can help management identify which product categories perform best in each branch and where improvement opportunities may exist. For example, the strong Food and beverages performance in Giza suggests high demand for this category in that branch. In contrast, the relatively low Health and beauty sales in Alex may justify further investigation into pricing, stock availability, promotions, or customer preferences.

8. Conclusion

The OLAP cube successfully transforms detailed sales transactions into a multidimensional summary that is easy to analyse. By using Sales as the measure and Branch and Product Line as dimensions, the analysis provides clear comparisons across locations and categories. Giza is the highest-performing branch overall, while Food and beverages is the strongest product line across all branches. This demonstrates how OLAP techniques can support faster business analysis and data-driven decision-making.

Source: Supplied Jupyter Notebook — FactSales, dimension tables, and OLAP cube output.